

Student Learning Outcomes of AKU-EB HSSC Computer Science Syllabus

Part I (Grade XI)

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level ¹		
			R	U	A and beyond
1. Computing Systems	Students should be able to:				
1.1 Data Representation in a Digital Computer	1.1.1	differentiate between analog and digital signals;		*	
	1.1.2	explain the binary data representation using binary pulses, i.e., 0/ low/ off and 1/ high/ on;		*	
1.2 Logic Gates	1.2.1	define the following terms: a. digital logic, b. logic gates and logic circuits, c. truth table;	*		
	1.2.2	explain the following logic gates in terms of the number of inputs and outputs using truth tables: a. AND, b. OR, c. NOT, d. NAND, e. NOR, f. Exclusive OR (XOR), g. Exclusive NOR (XNOR);		*	
	1.2.3	identify logic gates from the truth table;		*	
	1.2.4	explain the uses of logic gates in digital devices;		*	
	1.2.5	represent the logic circuits in the form of a truth table;			A

¹ R = Remember, U = Understand, A = Apply and beyond [Apply (A), Analyse (An), Evaluate (E), Create (C)]

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
			R	U	A and beyond
	Students should be able to:				
	1.2.6	explain the following Boolean identities: a. Identity Law, b. Distributive Law, c. Associative Law, d. Commutative Law, e. Inverse (Complement) Law, f. De Morgan's Theorem, g. Absorption Law;		*	
	1.2.7	construct a logic circuit for a given real-life problem;			A
1.3 Karnaugh Map (K-Map)	1.3.1	simplify two-variable and three-variable Boolean functions using Karnaugh map;			A
	1.3.2	convert the given algebraic expression into its Sum of Products (SOP) and Product of Sums (POS) forms;			A
1.4 Software Development	1.4.1	explain Software Development Life Cycle (SDLC) and its different phases;		*	
	1.4.2	compare the phases, advantages and disadvantages of the following software development models: a. waterfall model, b. agile model;		*	

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
			R	U	A and beyond
	Students should be able to:				
1.5 Network Topology	1.5.1	define network topology;	*		
	1.5.2	explain the following network topologies with diagrams: a. bus, b. ring, c. star, d. tree, e. mesh, f. hybrid;		*	
	1.5.3	explain the advantages, disadvantages and applications of network topologies mentioned in SLO # 1.5.2;		*	
	1.5.4	compare scalability and reliability of network topologies;		*	
1.6 Cloud Computing	1.6.1	explain cloud computing, its characteristics, and impact on modern IT infrastructure;		*	
	1.6.2	explain the following cloud computing service models with examples: a. Software as a Service (SaaS), b. Platform as a Service (PaaS), c. Infrastructure as a Service (IaaS);		*	
	1.6.3	explain the following cloud computing types: a. public cloud, b. private cloud, c. community cloud, d. hybrid cloud;		*	

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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	Students should be able to:				
1.7 Understanding Cybersecurity Fundamentals	1.7.1 1.7.2	describe cybersecurity and its importance; explain the following cybersecurity fundamentals: a. confidentiality, b. integrity, c. availability;		* *	
1.8 Common Cybersecurity Threats	1.8.1 1.8.2 1.8.3 1.8.4	explain the following cybersecurity threats and their impact: a. malware, b. phishing, c. pharming, d. Denial of Service (DoS), e. Distributed Denial of Service (DDoS), f. ransomware; recommend effective measures to safeguard against cybersecurity threats to enhance digital security; design a multi-layered security strategy for a given situation that integrates preventive, detective, and corrective measures to mitigate the risks associated with DoS, DDoS, and ransomware attacks; create a cybersecurity awareness campaign highlighting common threats and effective prevention methods for your community;		* <	

² FA= Formative Assessment, not to be assessed under examination conditions

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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2. Computational Thinking and Algorithm	Students should be able to:				
2.1 Computational Thinking and Artefacts	2.1.1 2.1.2 2.1.3	define the term computational thinking; describe different steps of computational thinking; explain the following computational artefacts of software development process: a. computational solution design, b. planning and development of computational artefacts, c. testing computational artefacts;	*	* *	
2.2 Algorithm	2.2.1	apply basic computational algorithms to perform the following operations in programming: a. arithmetic, relational, and logical computations, b. conditional (selection) statements, c. iterative (loop-based) processes, d. sorting techniques (insertion sort, bubble sort), e. searching techniques (binary search, linear search).			A

Topics and Sub-topics	Student Learning Outcomes	Cognitive Level			
		R	U	A and beyond	
3. Programming Fundamentals (Python)	Students should be able to:				
3.1 Programming Basics	3.1.1 define program and programming language; 3.1.2 differentiate between program syntax and program semantics; 3.1.3 explain the following types of programming languages: a. low-level languages: I. machine language, II. assembly language, b. high-level languages: I. procedural language, II. structured language, III. object-oriented language; 3.1.4 explain the functions of an/a assembler, compiler, and interpreter; 3.1.5 explain the Python language and its applications; 3.1.6 explain the Python Integrated Development Environment (IDE); 3.1.7 describe comments in Python programming; 3.1.8 write a Python program using single-line and multiple-line comments;	*	* * * * * *	A	

Topics and Sub-topics	Student Learning Outcomes	Cognitive Level			
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	Students should be able to:				
3.2 Turtle Graphics	3.2.1 define turtle graphics; 3.2.2 use turtle library in Python program; 3.2.3 draw different shapes using the following functions: a. Movement: get screen(), right() / rt(), left()/ lt(), forward()/ fd(), backward()/ bk(), b. Screen Functions: clear(), reset(), stamp(), clearstamp(), bgcolor(), title(), c. Changing Turtle Attributes: shapsize(), pensize(), fillcolor(), pencolor(), color(), shape(), speed(), begin_fill()....end_fill(), penup(), pendown(), d. Preset Shapes: circle(), dot();	*		A A	
3.3 Libraries	3.3.1 describe the purpose of libraries; 3.3.2 write the syntax of installing a library in Python language; 3.3.3 write a Python program that imports the datetime module and uses the following arguments: a. %A, b. %B, c. %C, d. %D, e. %H, f. %I, g. %S, h. %Y;		*	A A	

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		Students should be able to:					
3.4	Constants and Variables in Python Programming	3.4.1	define the following basic data types in Python programming: a. char, b. str, c. int, d. float, e. bool;	*			
		3.4.2	differentiate between variable and constant;		*		
		3.4.3	differentiate between local and global variables;		*		
		3.4.4	write valid variable names based on the variable naming rules;			A	
		3.4.5	write a Python program to a. utilise different data types, b. demonstrate implicit and explicit type casting of variables;			A	
3.5	Input Output Handling	3.5.1	describe the purpose of using eval() in Python;		*		
		3.5.2	write a Python program to use the eval() function;			A	
		3.5.3	write a program to display a message and the value of the variable using the print() statement;			A	
		3.5.4	write a program for taking input during the execution of a program using the input() statement;			A	

Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
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		Students should be able to:				
3.6	Operators in Python Programming	3.6.1	differentiate between operator and operand;		*	A
		3.6.2	describe the following types of operators: a. assignment operator, b. membership operators, c. arithmetic operators, d. bitwise operators, e. comparison (relational) operators, f. logical operators;		*	
		3.6.3	write a Python program using the following operators; a. arithmetic operators, b. bitwise operators, c. comparison (relational) operators, d. logical operators;			
3.7	Selection Statements in Python Programming	3.7.1	describe if , if-else , and elif statements;		*	A
		3.7.2	write a Python program using the following statements: a. if, b. if-else, c. if-elif-else, d. nested if, e. pass;			

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4. Data and Analysis	Students should be able to:				
4.1 Statistical Modelling	4.1.1	define the following terms: a. statistics, b. statistical modeling;	*		
	4.1.2	explain the following statistical modeling techniques (supervised and unsupervised): a. regression, b. classification, c. k-means clustering;		*	
	4.1.3	analyse the process of developing a statistical model in the context of data analysis;			An
	4.1.4	evaluate the effectiveness of statistical modeling in solving real-world problems in the fields of healthcare and finance;			E
4.2 Experimental Design in Data Science	4.2.1	define the following terms: a. correlation, b. causation, c. population, d. parameters, e. random sample;	*		
	4.2.2	differentiate between observational studies and experimental studies in data science;		*	
	4.2.3	describe the principles of experimental design flow;		*	
	4.2.4	explain the steps involved in the experimental design flow;		*	
4.3 Statistics and Data Visuals	4.3.1	explain the data collection methods and its types;		*	
	4.3.2	analyse the role of data science in addressing real-world problems, supported by practical examples.			An

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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5. Applications of Computer Science	Students should be able to:				
5.1 Introduction to Internet of Things (IoT)	5.1.1	describe Internet of Things (IoT) and its importance in connecting physical devices;		*	An
	5.1.2	explain the following components of an IoT system: a. sensors, b. processors, c. connectivity, d. user interface;		*	
	5.1.3	examine IoT's role in improving operations and addressing real-world challenges in manufacturing, agriculture, and healthcare sectors;			
5.2 Core Technologies Powering IoT	5.2.1	describe key technologies that enable Internet of Things (IoT);		*	
	5.2.2	describe big data analytics (BDA) and its applications in different fields;		*	
	5.2.3	describe communication protocols that facilitate data exchange between Internet of Things (IoT) devices;		*	
	5.2.4	explain embedded systems and their application;		*	
	5.2.5	explain Wireless Sensor Networks (WSN) and their applications in the Internet of Things (IoT);		*	
5.3 Blockchain and Blockchain Networks	5.3.1	define blockchain technology;	*		
	5.3.2	list the technologies that enable blockchain;	*		
	5.3.3	explain blockchain networks and the following types: a. public blockchain network, b. private blockchain network, c. permissioned blockchain network, d. consortium blockchain network;		*	
	5.3.4	explain the integration of blockchain and IoT;		*	

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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	Students should be able to:				
5.4 Stakeholders in AI Systems	5.4.1	describe the stakeholders involved in an Artificial Intelligence (AI) system;		*	
	5.4.2	evaluate the benefits and challenges of Artificial Intelligence (AI) in healthcare, education, and industry.			E

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6. Impacts of Computing	Students should be able to:				
6.1 Information Sources	6.1.1	define the following types of information sources: a. books, b. articles, c. websites;	*		
	6.1.2	explain reliable and unreliable sources of information;		*	
	6.1.3	explain the use of search engines and databases to find reliable information;		*	
	6.1.4	discuss the responsible and critical use of social media to access and share information;			E
6.2 Information Privacy and Security	6.2.1	define privacy and security;	*		
	6.2.2	describe the importance of protecting personal information online;		*	
	6.2.3	suggest strategies for safeguarding personal data while accessing information sources;			E
	6.2.4	evaluate the impact of data breaches and the misuse of information on individuals, organisations, and society;			E
6.3 Ethical Use of Information	6.3.1	explain the ethical use of information and intellectual property;		*	
	6.3.2	explain the consequences of misinformation and unethical information practices;		*	
	6.3.3	discuss real-world cases that involve ethical dilemmas related to the use of information and their effects on individuals, organisations, and society;			E

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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	Students should be able to:				
6.4 Connectivity in Computing	6.4.1	define the term 'connectivity' in the context of computing;	*		
	6.4.2	describe the following components involved in achieving connectivity: a. devices, b. networks, c. protocols, d. internet;		*	
	6.4.3	describe the following communication technologies enabled by computing: a. social media platforms, b. email, c. instant messaging, d. video conferencing;		*	
6.5 Environmental Impacts of Connectivity in Computing	6.5.1	explain the concept of green computing;		*	
	6.5.2	analyse the positive and negative environmental impacts of connectivity in computing across fields such as cloud computing and smart devices;			An
	6.5.3	examine the environmental impact of data centres, considering factors such as energy consumption and e-waste;			An
6.6 Cultural Impacts of Connectivity	6.6.1	analyse the role of enhanced connectivity in facilitating global cultural exchange;			An
	6.6.2	discuss the preservation or loss of local cultures in the digital age;			E
6.7 Human Impact of Connectivity	6.7.1	discuss the positive and negative impacts of digital connectivity on human behaviour and society, including work and employment.			E

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7. Digital Literacy	Students should be able to:				
7.1 Introduction to Digital Literacy	7.1.1	describe digital literacy and its importance;		*	
	7.1.2	explain the role of digital literacy in academic and everyday life;		*	
7.2 Introduction to Advanced Search Techniques	7.2.1	describe advanced search techniques and their importance;		*	
	7.2.2	differentiate between basic and advanced searches;		*	
7.3 Designing Data-Collection Approaches	7.3.1	describe the following data collection methods: a. qualitative, b. quantitative;		*	
	7.3.2	discuss the use of prototypes and simulations in data collection, including their purpose and advantages;			E
	7.3.3	design a data-collection plan using qualitative interviews by: a. formulating research questions, b. developing interview guides and protocols;			C
	7.3.4	create surveys for quantitative data collection by: a. writing clear and unbiased survey questions, b. selecting appropriate survey platforms and tools.			C

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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8. Entrepreneurship in the Digital Age	Students should be able to:				
8.1 Designing and Iterating User Centered Prototypes	8.1.1	apply basic design principles to develop a low-fidelity prototype that addresses a specific user problem;			FA
	8.1.2	illustrate the user journey and key features of the prototype using visual tools such as sketches or storyboards;			FA
	8.1.3	determine improvements in the prototype based on user testing and feedback;			FA
	8.1.4	analyse test results to identify whether the prototype meets user needs or validates a business assumption;			FA
	8.1.5	evaluate the effectiveness of the final prototype in addressing user needs and suggest future improvements.			FA

Student Learning Outcomes of AKU-EB HSSC Computer Science Syllabus

Grade XII

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level ¹		
			R	U	A and beyond
9. Computing Systems	Students should be able to:				
9.1 Usability, Security and Accessibility of Devices	9.1.1	describe the use of digital device in terms of ease of use and efficiency;		*	
	9.1.2	describe the security features of devices and best practices for data protection;		*	
	9.1.3	explain the ways in which accessibility features in devices support users with disabilities;		*	
9.2 Human Interaction with Computer Systems in Terms of Usability	9.2.1	describe Human-Computer Interaction (HCI) and its importance;		*	An
	9.2.2	analyse the influence of Human-Computer Interaction (HCI) on user efficiency and task effectiveness;			
9.3 Design Consideration	9.3.1	describe HCI design considerations and their effects on usability and system performance;		*	An
	9.3.2	analyse the trade-offs involved in designing HCI in the context of: a. security, b. user-friendliness, c. cost-effectiveness.			

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Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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10. Computational Thinking and Algorithm					
10.1 Complex Data Structure	10.1.1 10.1.2	define data structure; describe the following types of data structures: a. list, b. array, c. tree, d. stack, e. queue;	*	*	
10.2 Operations on Data Structures	10.2.1 10.2.2 10.2.3 10.2.4	describe the traversal operation on an array; describe the enqueue and dequeue operations in a queue; explain the push and pop operations in a stack; explain the following operations on a tree: a. traversing, I. in order, II. pre-order, III. post order, b. binary search;		* * * *	
10.3 Applications of Data Structures	10.3.1 10.3.2 10.3.3 10.3.4 10.3.5	explain the application of tree data structures; explain the application of the binary search algorithms; apply tree data structures to represent hierarchical data models such as file systems or organisational charts; use the binary search algorithm to locate elements in sorted data efficiently; illustrate the use of tree traversal techniques in data retrieval and manipulation tasks.		* *	A A A

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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11. Programming Fundamentals (Python)		Students should be able to:			
11.1 Introduction to Function in Python Programming	11.1.1 describe predefined and user-defined functions; 11.1.2 describe the advantages of user-defined functions; 11.1.3 describe the signature of the functions, i.e., function name, arguments, and return value; 11.1.4 explain the process of creating a user-defined function, i.e., function declaration, function definition, and function call; 11.1.5 differentiate among the following variable types used in functions: a. local, b. global; 11.1.6 write a program involving a user-defined function;		* * * * *	A	
11.2 Passing Arguments and Returning Values	11.2.1 define default argument; 11.2.2 describe the purpose of the return statement; 11.2.3 write a program that demonstrates the return statement; 11.2.4 write a program to invoke a user-defined function and pass arguments by constant, value, and reference;	*	*	A A	
11.3 Classes and Objects in Object-Oriented Programming (OOP)	11.3.1 describe Object-Oriented Programming (OOP) and functional programming; 11.3.2 compare the advantages and disadvantages of Object-Oriented Programming (OOP) and functional programming; 11.3.3 explain class and object in terms of OOP; 11.3.4 write a Python program to a. declare a class with data members and member functions defined within its body, b. create objects of the class in the main() function and call its member functions using those objects;		* * *	A	

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	Students should be able to:				
11.4 Access Modifiers (Public, Private and Protected)	11.4.1	differentiate between private and public access specifiers in Object-Oriented Programming (OOP);		*	
	11.4.2	write a Python program that demonstrates private access to class members using name mangling;			A
	11.4.3	write a Python program that demonstrates public access to class members;			A
	11.4.4	write a Python program that defines a class with both public and private members and demonstrates their usage;			A
11.5 Pillars of OOP (Inheritance, Encapsulation, Abstraction and Polymorphism)	11.5.1	describe inheritance and its access specifiers in Python programming;		*	
	11.5.2	explain the types of inheritance in Python programming using different classes;		*	
	11.5.3	explain encapsulation in Python programming;		*	
	11.5.4	explain polymorphism in Python programming;		*	
	11.5.5	explain the function of constructors and destructors in Object-Oriented Programming (OOP);		*	
	11.5.6	differentiate among the types of constructors, i.e., default constructor, user-defined constructor, and constructor overloading;		*	
	11.5.7	differentiate between overloading and overriding in OOP;		*	
	11.5.8	write a program of inheritance using the base class and the derived class;			A
	11.5.9	write a program of inheritance for three access specifiers;			A

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
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	Students should be able to:				
11.6 Working with List	11.6.1	explain the concept of a list in Python programming including its name, size, index number, and arrangement of elements in the memory;		*	
	11.6.2	explain the purpose of storing elements in an ordered list;		*	
	11.6.3	write a Python program to use a one-dimensional list with different sizes and data types;			A
	11.6.4	write a Python program to create, add, modify, and remove items in a list;			A
	11.6.5	write a Python program that stores numeric values in a one-dimensional list using a for loop and find the highest, lowest, average, and search for a value;			A
11.7 Working with Dictionaries	11.7.1	describe dictionary and its uses in Python programming;		*	
	11.7.2	write a Python program to construct and retrieve values from a dictionary in Python programming;			A
	11.7.3	write a Python program to find a value in a dictionary using the key value;			A
11.8 File Handling in Python Programming	11.8.1	describe the importance of disk I/O;		*	
	11.8.2	explain file handling methods and the following file operations in Python programming: a. create, b. read, c. write, d. append;		*	
	11.8.3	write a Python program that performs the following file operations: a. create and write data to a file, b. read from and append data to the file, c. write new data and then read it from the file;			A

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	Students should be able to:			
11.9 Introduction to Database (MS SQL)	11.9.1 describe database and its importance; 11.9.2 differentiate between Database Management System (DBMS) and Relational Database Management System (RDBMS); 11.9.3 describe the MS SQL database; 11.9.4 describe the following keys: a. primary key, b. foreign key, c. candidate key, d. alternate key e. secondary key; 11.9.5 construct a database and tables using MS SQL;		* * * *	A
11.10 Data Normalisation	11.10.1 describe data normalisation and its importance in a database; 11.10.2 explain 1NF, 2NF and 3NF forms of data normalisation;		* *	
11.11 Structure Query Language (SQL)	11.11.1 describe the Structured Query Language (SQL) and its importance; 11.11.2 explain the following types of SQL statements: a. Data Manipulation Language (DML), b. Data Definition Language (DDL), c. Data Control Language (DCL), d. Transaction Control Language (TCL), e. Data Query Language (DQL); 11.11.3 explain data manipulation language (DML) statements ; 11.11.4 apply SELECT statement with clauses: a. WHERE, b. Order by;		* * *	A
11.12 Connecting Database with Python	11.12.1 explain the process of connecting a database with Python; 11.12.2 write a program to connect a database with Python.		*	A

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12. Data and Analysis		Students should be able to:				
12.1	Introduction to Machine Learning (ML)	12.1.1	define Machine Learning (ML);	*		
		12.1.2	differentiate between Machine Learning (ML) and rule-based algorithm;		*	
		12.1.3	explain the following basics of Machine Learning (ML) model building: a. feature engineering, b. train test split, c. model assessment;		*	
		12.1.4	explain the following Machine Learning (ML) algorithms: a. regression, b. clustering, c. classification;		*	
12.2	Hypothesis and Hypothesis Testing	12.2.1	describe hypothesis and its role in statistical analysis;		*	
		12.2.2	describe the null hypothesis and the alternative hypothesis;		*	
		12.2.3	explain P value and significance testing;		*	
		12.2.4	explain the following tests: a. f-test, b. chi-square test, c. ANOVA;		*	
12.3	Key Model Performance Metrics	12.3.1	explain the following performance evaluation techniques; a. accuracy, b. precision, c. recall, d. f1 score, e. rand index, f. ROC (Receiver Operating Characteristic), g. RMSE (Root Mean Square Error).		*	

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13. Applications of Computer Science	Students should be able to:				
13.1 Applications of IoT (Internet of Things), Cloud Computing and Blockchain in Pakistan	13.1.1	describe the following key applications of IoT in Pakistan: a. smart homes, b. agriculture, c. smart cities, d. healthcare monitoring systems;		*	
	13.1.2	explain the importance of cloud computing in Pakistan for data storage;		*	
	13.1.3	evaluate the benefits and challenges of cloud computing for data management in Pakistan;			E
13.2 Deep Learning and its Application	13.2.1	explain the relationship between deep learning networks and neural networks;		*	
	13.2.2	analyse the applications of neural networks, including deep learning models, in fields such as healthcare, finance, and autonomous vehicles.			An

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14. Impacts of Computing	Students should be able to:				
14.1 Introduction to Privacy and Security	14.1.1	define the terms, privacy and security;	*		
	14.1.2	describe the following common types of cyberattacks: a. Denial of Service (DoS) and Distributed Denial of Service (DDoS) attack, b. ransomware, c. spyware, d. viruses, e. phishing, f. Domain Name System (DNS) spoofing, g. waterhole attack, h. fake Wireless Access Point (WAP), i. eavesdropping;		*	
	14.1.3	explain the following basic security methods: a. strong passwords, b. Two-Factor Authentication (2FA), c. antivirus and anti-malware software, d. checking logs, e. firewall, f. Intrusion Detection System (IDS); g. encryption;		*	

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	Students should be able to:				
14.2 Security Protocols	14.2.1	explain the following network security protocols: a. Secure Socket Layer (SSL) b. Transport Layer Security (TLS), c. Internet Protocol Security (IPsec), d. Secure Shell (SSH), e. Hypertext Transfer Protocol (HTTP), f. Hypertext Transfer Protocol Secure (HTTPS), g. Wireless Protected Access (WPA/ WPA2/ WPA3), h. File Transfer Protocol Secure (FTPS), i. Simple Network Management Protocol (SNMP);		*	
	14.2.2	differentiate between HTTP and HTTPS;		*	
14.3 Cryptography	14.3.1	define cryptography;	*		
	14.3.2	differentiate between cybersecurity and cryptography;		*	
	14.3.3	explain the following different types of ciphers: a. plain, b. ceaser, c. shift, d. substitution, e. transposition, f. stream cipher, g. block cipher.		*	

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15. Digital Literacy		Students should be able to:					
15.1	Understanding Advance Searches and Digital Tools in Research	15.1.1	define research questions/ problems;	*			
		15.1.2	describe a case study in research;		*		
		15.1.3	analyse the effectiveness of digital tools in communicating research results and conclusions;			An	
		15.1.4	evaluate the effectiveness of advanced search techniques in locating information on a research topic;			E	
15.2	Plan, Execute, and Communicate Research	15.2.1	design a data collection plan to answer a research question and present the data using appropriate digital tools.			FA ²	

²FA= Formative Assessment, not to be assessed under examination conditions

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level			
			R	U	A and beyond	
16. Entrepreneurship in the Digital Age	Students should be able to:					
16.1 Understanding Minimum Viable Product (MVP) and Business Ideas	16.1.1	define Minimum Viable Product (MVP);	*			
	16.1.2	explain the key characteristics of a successful MVP;				
	16.1.3	differentiate between MVP and prototype;			*	
	16.1.4	explain the riskiest assumption (Business Model);			*	
	16.1.5	apply the principles of MVP development to design a simple prototype for a specific user need;				A
	16.1.6	analyse the process of designing a test using an MVP;				An
	16.1.7	analyse case studies of successful businesses that created their MVPs.				An

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